



Operations Research

05 – Transportation model

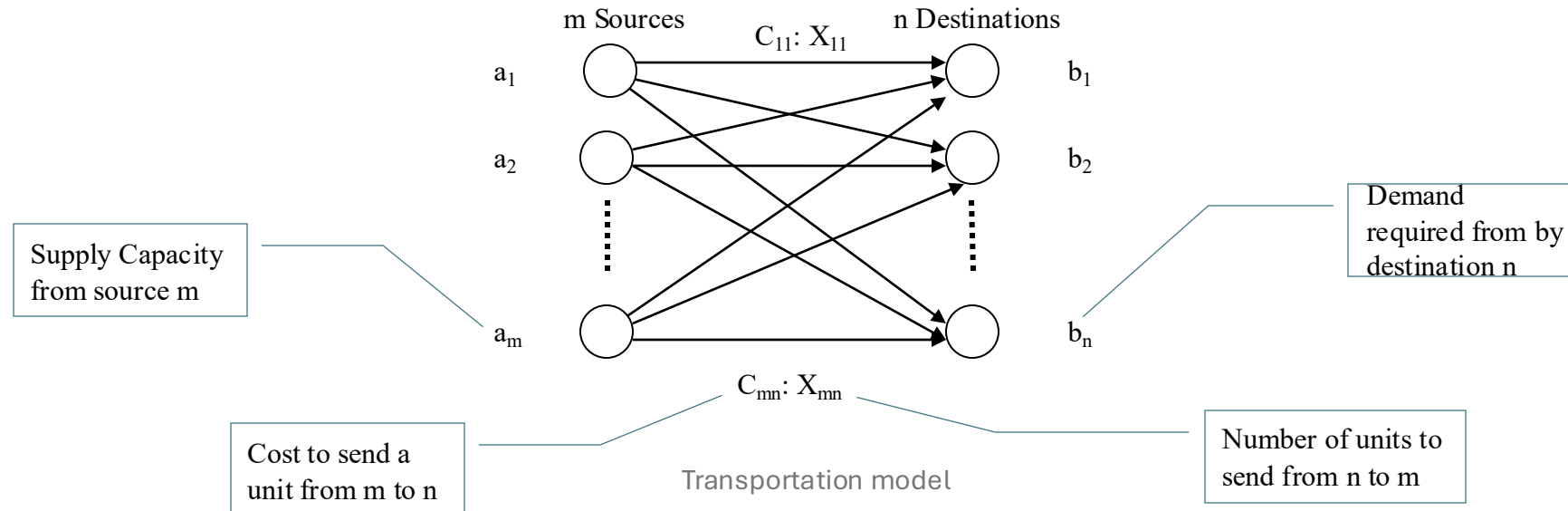
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Objectives

- Given a situation, identify when the transportation algorithm can be applied
- Understand the basics of the transportation algorithm
- Generate a basic feasible solution for the transportation problem

The Transportation Problem

- The transportation problem is a special case of an LP problem
- Because of its characteristics, the transportation problem can be solved very efficiently with a special algorithm called the transportation algorithm
- The problem is concerned with specifying how to disposition a single product from several sources to several destinations at minimum cost



Car-Distribution Problem (from Taha)

The MG Auto Company has plants in LA, Detroit, and New Orleans. Its major distribution centers are in Denver and Miami. The capacities of the three plants during the next quarter are: 1000, 1500, and 1200 cars. The quarterly demands at the two distribution centers are 2300 and 1400 cars. The cost of shipping in \$'s per car is given by:

	Denver	Miami
Los Angeles	80	215
Detroit	100	108
New Orleans	102	68

Find the best strategy to send cars from the plants to the distribution centers.

Model of the Car Distribution Problem

Example: The car distribution problem

Let X_{ij} = units sent from location i (L,N,D) to destination j (V,M).

Then, the problem can be stated as:

$$\text{Minimize } z = 80X_{LV} + 215X_{LM} + 100X_{DV} + 108X_{DM} + 102X_{NV} + 68X_{NM}$$

Subject to:

$$X_{LV} + X_{LM} = 1000$$

$$X_{DV} + X_{DM} = 1500$$

$$X_{NV} + X_{NM} = 1200$$

$$X_{LV} + X_{DV} + X_{NV} = 2300$$

$$X_{LM} + X_{DM} + X_{NM} = 1400$$

$$X_{LV}, X_{LM}, X_{DV}, X_{DM}, X_{NV}, X_{NM} \geq 0$$

The General Transportation Problem

- In general, a transportation problem can be expressed as:

$$\min \quad z = \sum_{i=1}^m \sum_{j=1}^n C_{ij} X_{ij}$$

Subject to:

$$\sum_{j=1}^n X_{ij} = a_i, i = 1, 2, \dots, m$$

$$\sum_{i=1}^m X_{ij} = b_j, j = 1, 2, \dots, n$$

$$X_{ij} \geq 0$$

Transportation Tableau

- Usually, it is not necessary to explicitly build the LP model of the transportation problem. Instead, we usually represent the transportation problem by using a transportation tableau

Decision Variable

		Markets (Destinations)				
		M_1	M_2	M_3	M_4	
Sources	S_1	X_{11} C_{11}	X_{12} C_{12}	X_{13} C_{13}	X_{14} C_{14}	a_1
	S_2	X_{21} C_{21}	X_{22} C_{22}	X_{23} C_{23}	X_{24} C_{24}	a_2
	S_3	X_{31} C_{31}	X_{32} C_{32}	X_{33} C_{33}	X_{34} C_{34}	a_3
		b_1	b_2	b_3	b_4	

Number of units required in destination 1 (j)

Cost for sending a unit from 3 (i) to 1 (j)

Number of available units to ship from source 3 (i)

Transportation Tableau Example

- Example: Car Distribution Matrix

		Markets (Destinations)		Supply
		Denver	Miami	
Plant	L.A.	X_{11} 80	X_{12} 215	1000
	Detroit	X_{21} 100	X_{22} 108	1500
	New Orl.	X_{31} 102	X_{32} 68	1200
	Demand	2300	1400	

$$\text{Min } z = 80 X_{11} + 215 X_{12} + 100 X_{21} + 108 X_{22} + 102 X_{31} + 68 X_{32}$$

$$X_{11} + X_{12} = 1000 \text{ (LA)}$$

$$X_{21} + X_{22} = 1500 \text{ (Detroit)}$$

$$X_{31} + X_{32} = 1200 \text{ (New O.)}$$

$$X_{11} + X_{21} + X_{31} = 2300 \text{ (Denver)}$$

$$+ X_{12} + X_{22} + X_{32} = 1400 \text{ (Miami)}$$

Characteristics of the Transportation Problem

- The transportation problem could be solved using the regular simplex method. However, because of its special characteristics, a more efficient procedure is used. The procedure is called the transportation (simplex) method.
- Because of the uni-modularity property (special structure of the constraints), transportation problems with supplies and demands given by integers will have integer basic solutions.
- The transportation problem is solved in two phases:
 1. Determination of an initial basic feasible solution
 2. Finding an optimal solution through the sequential improvement of the initial feasible solution

Finding an Initial Basic Solution

- There are many ways to find a feasible solution. We will examine several below. The first is simple but ineffective, and we will then look at more complex but effective (producing near optimal solutions) methods.
- **Northwest Corner Rule (NWC)**
 - We begin in the Northwest (upper-left) corner of the matrix and assign as much as we can (considering supply and demand) and update remaining supply and demand. We move either down or to the right (depending on whether supply or demand has been depleted). We again assign as much as possible and continue to the Southeast (lower-right).

Northwest Corner Rule (NWC)

- **Step1.** Allocate as much as possible to the selected cell and adjust the associated amounts of supply and demand by subtracting the allocated amount.
- **Step2.** Cross out the row or column with zero supply or demand. If both a row and a column net to zero simultaneously, cross out one only, and leave a zero supply (demand) in the uncrossed-out row (column).
- **Step3.** If exactly one row or column is left uncrossed out, stop. Otherwise, move to the cell to the right if a column has just been crossed out or below if a row has been crossed out. Go step 1

Example of NW Corner

		Markets (Destinations)		Supply
		Denver	Miami	
Plant	L.A.	X_{11} 80	X_{12} 215	1000
	Detroit	X_{21} 100	X_{22} 108	1500
	New Orl.	X_{31} 102	X_{32} 68	1200
	Demand	2300	1400	

		Markets (Destinations)		Supply
		Denver	Miami	
Plant	L.A.	1000 ↓	215	1000
	Detroit	1300 ↓	200 →	1500
	New Orl.	102	1200 ↓	1200
	Demand	2300	1400	

TC=313,200

Example 2 of NWC

- Using the NW corner rule, make the initial assignment for the following transportation problem

	Demand				
Source	A	B	C	D	
1	X_{11} 10	X_{12} 2	X_{13} 20	X_{14} 11	15
2	X_{21} 12	X_{22} 7	X_{23} 9	X_{24} 20	25
3	X_{31} 4	X_{32} 14	X_{33} 16	X_{34} 18	10
	5	15	15	15	

- the starting basic solution are:

$$\begin{aligned}
 X_{11} &= 5, & X_{12} &= 10; \\
 X_{22} &= 5, & X_{23} &= 15, & X_{24} &= 5 \\
 X_{34} &= 10
 \end{aligned}$$

- the association cost is

$$z = 5 \cdot 10 + 10 \cdot 2 + 5 \cdot 7 + 15 \cdot 9 + 5 \cdot 20 + 10 \cdot 18 = \$520$$

The Least Cost Rule (LCR)

- Note that the NWC method did not look at the costs! Thus, it may produce a terrible solution.
- The Least Cost Rule examines the costs to build an initial solution.
- The cell with the lowest cost is chosen, and we assign as many units as possible to the cell (considering supply and demand).
- We then reduce supplies and demands by the assignment and mark out ineligible cells (those in rows or columns where the supply or demand has been depleted).
- We repeat this process until all supplies and demands are depleted.

Example LCR

		Markets (Destinations)		Supply
		Denver	Miami	
Plant	L.A.	1000		1000
		80	215	0
	Detroit	1300	200	1500
		100	108	200 0
	New Orl.		1200	1200
		102	68	0
	Demand	2300	1400	
		1300	200 0	
		0		

Least Cost

Least Cost

TC=313,200

Example 2 of LCR

- Using the LC Rule, make the initial assignment for the following Transportation Problem

	Demand				
Source	A	B	C	D	
1	X_{11} 10	X_{12} 2	X_{13} 20	X_{14} 11	15
2	X_{21} 12	X_{22} 7	X_{23} 9	X_{24} 20	25
3	X_{31} 4	X_{32} 14	X_{33} 16	X_{34} 18	10
	5	15	15	15	0

Least
Cost

Least Cost

Least Cost

Least Cost

Least Cost

Least Cost

Example 2 of LCR (cont.)

	Demand				
Source	A	B	C	D	
1	X_{11} 10	X_{12} 2	X_{13} 20	X_{14} 11	15
2	X_{21} 12	X_{22} 7	X_{23} 9	X_{24} 20	25
3	X_{31} 4	X_{32} 14	X_{33} 16	X_{34} 18	10
	5	15	15	15	

- the starting basic solution are:

$X_{12}=15$; $X_{14}=0$;
 $X_{23}=15$; $X_{24}=10$;
 $X_{31}=5$ $X_{33}=5$

- the association cost is

$$z = 15 \cdot 2 + 0 \cdot 11 + 15 \cdot 9 + 10 \cdot 20 + 5 \cdot 4 + 5 \cdot 18 = \$475$$

Vogel's Approximation Method (VAM)

- This method recognizes that it may be wise to make a small sacrifice for a bigger gain. It computes a penalty for each row and column if the lowest cost cell is not selected. That is, it figures out what it would cost to take the second-best cost.
- The penalty is the cost difference between the lowest cost cell and next lowest cost value in each row and column.
- We select the cell associated with the largest penalty to assign units to and proceed essentially like the LCR. We will have to recalculate some of the penalties on each iteration.

Vogel's Approximation Method (VAM)

- **Step 1:** Determine the difference between the lowest two cells in all rows and columns, including dummies.
- **Step 2:** Identify the row or column with the largest difference. Ties may be broken arbitrarily.
- **Step 3:** Allocate as much as possible to the lowest-cost cell in the row or column with the highest difference. If two or more differences are equal, allocate as much as possible to the lowest-cost cell in these rows or columns.
- **Step 4:** Stop the process if all row and column requirements are met. If not, go to the next step.
- **Step 5:** Recalculate the differences between the two lowest cells remaining in all rows and columns. Any row and column with zero supply or demand should not be used in calculating further differences. Then go to Step 2.

VAM

When this assignment is made, we deplete both the column and the row; however, we eliminate just one. In this case, we arbitrarily eliminate the row

	Demand					Penalty
Source	A	B	C	D	Avail	
1	10	15	20	11	15	10
2	0	0	15	10	25	2
3	5				5	14
Demand	5	15	15	10		

High
est
Penal
ty

When this assignment is made, we deplete both the column and the row; however, we eliminate just one. In this case, we arbitrarily eliminate the row

Highest
Penalty.

VAM Final Solution

	Demand				
Source	A	B	C	D	Avail
1	10	15 0	20	11	15
2	0 12	0 7	15 9	10 20	25
3	5 0	14	16	18	5
Demand	5	15	15	10	

TC = 335

Example 2 of VAM

	Demand				
Source	A	B	C	D	Avail
1	10	2	20	11	15
2	12	7	9	20	25
3	4	14	16	18	10
Demand	5	15	15	10	

Penalty	High est Penal
8	9
2	11
10	2

- the starting basic solution are:

$$\begin{aligned} X_{12} &= 15; & X_{14} &= 0; \\ X_{23} &= 15; & X_{24} &= 10; \\ X_{31} &= 5 & X_{33} &= 5 \end{aligned}$$

- the association cost is

$$z = 15 \cdot 2 + 0 \cdot 11 + 15 \cdot 9 + 10 \cdot 20 + 5 \cdot 4 + 5 \cdot 18 = \$475$$

The solution happens to have the same objective value as in LCR

Dealing with Unbalanced Problems

- If the problem is unbalanced (demand and supply are not equal), the problem can be transformed into a balanced one by creating dummy sources or destinations with a cost of zero (usually). These dummy nodes will absorb the difference between the supply and demand.

Dealing with Unbalanced Problems

- Balanced: total demand = total supply
- Example: We have three reservoirs with daily supplies of 15, 20, and 25 million liters of fresh water, respectively. On each day we must supply four cities- A, B, C and D, whose demands are 8, 10, 12, and 15, respectively. The cost of pumping per million liters is given below:

	Cities			
Reservoirs	A	B	C	D
1	2	3	4	5
2	3	2	5	2
3	4	1	2	3

- Set up the transportation Tableau to determine the cheapest pumping schedule if excess water can be disposed of at no cost.

	City					
Reservoir	A	B	C	D	E (Dummy)	
1	X_{11} 2	X_{12} 3	X_{13} 4	X_{14} 5	X_{15} 0	15
2	X_{21} 3	X_{22} 2	X_{23} 5	X_{24} 2	X_{25} 0	20
3	X_{31} 4	X_{32} 1	X_{33} 2	X_{34} 3	X_{35} 0	25
Demand	8	10	12	15	15	

Dummy
Destination
Added

Tutorial

1. Compare the starting solutions obtained by the northwest-corner, least-cost, and Vogel methods for each of the following models:

***(a)**

0	2	1	6
2	1	5	7
2	4	3	7
5	5	10	

(b)

1	2	6	7
0	4	2	12
3	1	5	11
10	10	10	

(c)

5	1	8	12
2	4	0	14
3	6	7	4
9	10	11	

Iterative computation of the transportation algorithms

- After determining the starting solution; use the following algorithms to determine the optimum solution

Step1 : use the simplex optimality condition to determine the Entering Variables as the current nonbasic variable that can improve the solution. If the optimality condition is satisfied, stop. Otherwise, go to step 2

Step 2: determine the Leaving variables using the simplex feasibility condition. Change the basis and return to step 1.

- The optimality and feasibility do not involve the row operational that used in simplex method. Instead, the special structure of transportation allow simpler computation

Iterative computation of the transportation algorithms

- The determination of the Entering variable is done by computing the nonbasic coefficient in z-row , using **Method of multipliers**
- In the method of multiplier, associate the multipliers **u_i** and **v_j** with **row i** and **column j** of the transportation tubule.
- These multipliers satisfy the following equations:
$$u_i + v_j = C_{ij}, \text{ for each basic } X_{ij}$$
- To solve these equation, the method of multipliers call for arbitrarily setting any $u_i=0$, and then solving for the remaining variables

Example

- By using the NW corner rule, the starting basic solution are:

	Demand				
Source	A	B	C	D	
1	X_{11} 10 5	X_{12} 2 10	X_{13} 20	X_{14} 11	15
2	X_{21} 12	X_{22} 7 5	X_{23} 9 15	X_{24} 20 5	25
3	X_{31} 4	X_{32} 14	X_{33} 16	X_{34} 18 10	10
	5	15	15	15	

Example

	Demand				
Source	A	B	C	D	
1	X_{11} 5 10	X_{12} 10 2	X_{13} 20	X_{14} 11	15
2	X_{21} 12	X_{22} 5 7	X_{23} 15 9	X_{24} 5 20	25
3	X_{31} 4	X_{32} 14	X_{33} 16	X_{34} 10 18	10
	5	15	15	15	

Basic variable	(u,v) Equation	Solution
X11	$U_1 + V_1 = 10$	Set $U_1=0 \rightarrow V_1=10$
X12	$U_1 + V_2 = 2$	$U_1=0 \rightarrow V_2=2$
X22	$U_2 + V_2 = 7$	$V_2=2 \rightarrow U_2=5$
X23	$U_2 + V_3 = 9$	$U_2=5 \rightarrow V_3=4$
X24	$U_2 + V_4 = 20$	$U_2=5 \rightarrow V_4=15$
X34	$U_3 + V_4 = 18$	$V_4=15 \rightarrow U_3=3$

**We got: $U_1=0, U_2=5, U_3=3$
 $V_1=10, V_2=2, V_3=4, V_4=15$**

Example

- Use u_i , v_j to evaluate the nonbasic variable by computing:

$U_i + v_j - c_{ij}$, for each nonbasic x_{ij}

	Demand				
Source	A	B	C	D	
1	X_{11} 5 10	X_{12} 10 2	X_{13} 20	X_{14} 11	15
2	X_{21} 12	X_{22} 5 7	X_{23} 15 9	X_{24} 5 20	25
3	X_{31} 4	X_{32} 14	X_{33} 16	X_{34} 10 18	10
	5	15	15	15	

Nonbasic variable	$U_i + V_j - c_{ij}$
X13	$U_1 + V_3 - C_{13} = 0 + 4 - 20 = -16$
X14	$U_1 + V_4 - C_{14} = 0 + 15 - 11 = 4$
X21	$U_2 + V_1 - C_{21} = 5 + 10 - 12 = 3$
X31	$U_3 + V_1 - C_{31} = 3 + 10 - 4 = 9$
X32	$U_3 + V_2 - C_{32} = 3 + 2 - 14 = -9$
X33	$U_3 + V_3 - C_{33} = 3 + 4 - 16 = -9$

Example

- $u_i + v_j - c_{ij} = 0$ for each basic x_{ij} is important to computing the z-row of the simplex tableau, as the following:

Basic	X11	X12	X13	X14	X21	X22	X23	X24	X31	X32	X33	X34
Z	0	0	-16	4	3	0	0	0	9	-9	-9	0

- The transportation seek to minimize so, EV is the largest positive value coefficient in z-row thus X31 is EV

Basic	X11	X12	X13	X14	X21	X22	X23	X24	X31	X32	X33	X34
Z	0	0	-16	4	3	0	0	0	9	-9	-9	0

Example

- Construct closed loop that start and end at EV;
 - The loop consist of connected horizontal and vertical segments only.
 - Except EV, each corner must coincide with basic variable.
 - It must alternate between subtracting and adding (an addition to one cell in the loop is followed by a subtraction from the next cell in the loop)

	Demand				
Source	A	B	C	D	
1	$X_{11} - 5$ 10	$X_{12} + 10$ 2	X_{13} 20	X_{14} 11	15
2	X_{21} 12	$X_{22} - 5$ 7	$X_{23} + 15$ 9	$X_{24} + 5$ 20	25
3	$X_{31} + 4$ 4	X_{32} 14	X_{33} 16	$X_{34} - 10$ 18	10
	5	15	15	15	

each unit shipped here will save \$9

Example

- To determine the LV it should Determine Which Current Basic Variable Reaches 0 First.
 - If we add one unit to X_{31} , it must subtract a unit from X_{34} (leaving 9); add it back to X_{24} (giving 6), and subtract it from X_{22} (leaving 4); add it back to X_{12} (giving 11), and subtract it from X_{11} (leaving 4); this saves $(1)(\$9)$.
 - If we add five unit to X_{31} , it must subtract a 5 unit from X_{34} (leaving 5); add it back to X_{24} (giving 10), and subtract it from X_{22} (leaving 0); add it back to X_{12} (giving 15), and subtract it from X_{11} (leaving 0); this saves $(5)(\$9) = \45 .

	Demand				
Source	A	B	C	D	
1	$X_{11} - 5$ 10	$X_{12} + 10$ 2	X_{13} 20	X_{14} 11	15
2	X_{21} 12	$X_{22} 5$ 7	$X_{23} 15$ 9	$X_{24} 5$ 20	25
3	$X_{31} + 4$ 4	X_{32} 14	X_{33} 16	$X_{34} 10$ 18	10
	5	15	15	15	

Both X_{11} , X_{22} reach zero, arbitrary choose X_{11} to leave the solution. The new cost is $520 - 45 = \$475$

Example

- The next transportation tableau is

	Demand				
Source	A	B	C	D	
1	X_{11} 10	X_{12} 15 2	X_{13} 20	X_{14} 11	15
2	X_{21} 12	X_{22} 0 7	X_{23} 15 9	X_{24} 10 20	25
3	X_{31} 5 4	X_{32} 14	X_{33} 16	X_{34} 5 18	10
	5	15	15	15	

Basic variable	(u,v) Equation	Solution
X12	$U_1 + V_2 = 2$	Set $U_1=0 \rightarrow V_2=2$
X22	$U_2 + V_2 = 7$	$V_2=2 \rightarrow U_2=5$
X23	$U_2 + V_3 = 9$	$U_2=5 \rightarrow V_3=4$
X24	$U_2 + V_4 = 20$	$U_2=5 \rightarrow V_4=15$
X34	$U_3 + V_4 = 18$	$V_4=15 \rightarrow U_3=3$
X31	$U_3 + V_1 = 4$	$U_3=3 \rightarrow V_1=1$

Example

- Use u_i , v_j to evaluate the nonbasic variable by computing:

$U_i + v_j - c_{ij}$, for each nonbasic x_{ij}

	Demand				
Source	A	B	C	D	
1	X_{11} 10	X_{12} 15 2	X_{13} 20	X_{14} 11	15
2	X_{21} 12	X_{22} 0 7	X_{23} 15 9	X_{24} 10 20	25
3	X_{31} 5 4	X_{32} 14	X_{33} 16	X_{34} 5 18	10
	5	15	15	15	

Nonbasic variable	$U_i + V_j - c_{ij}$
X_{11}	$U_1 + V_1 - C_{11} = 0 + 1 - 10 = -9$
X_{13}	$U_1 + V_3 - C_{13} = 0 + 4 - 20 = -16$
X_{14}	$U_1 + V_4 - C_{14} = 0 + 15 - 11 = 4$
X_{21}	$U_2 + V_1 - C_{21} = 5 + 1 - 12 = -6$
X_{32}	$U_3 + V_2 - C_{32} = 3 + 2 - 14 = -9$
X_{33}	$U_3 + V_3 - C_{33} = 3 + 4 - 16 = -9$

Example

- EV is

Basic	X11	X12	X13	X14	X21	X22	X23	X24	X31	X32	X33	X34
Z	-9	0	-16	4	-6	0	0	0	0	-9	-9	0

- closed loop

	Demand				
Source	A	B	C	D	
1	X ₁₁ 10	X ₁₂ 2	X ₁₃ 20	X ₁₄ 11	15
2	X ₂₁ 12	X ₂₂ 7	X ₂₃ 9	X ₂₄ 20	25
3	X ₃₁ 4	X ₃₂ 14	X ₃₃ 16	X ₃₄ 18	10
	5	15	15	15	

- If we add 10 unit to X₁₄, it must subtract a 10 unit from X₂₄ (leaving 0); add it back to X₂₂ (giving 10), and subtract it from X₁₂ (leaving 5; this saves (10)(\$4)= \$45.

X₂₄ reach zero, so it leaves the solution

The new cost is 475 - 40 = \$435

Example

- The next transportation tableau is

	Demand				
Source	A	B	C	D	
1	X_{11} 10	X_{12} 5 2	X_{13} 20	X_{14} 10 11	15
2	X_{21} 12	X_{22} 10 7	X_{23} 15 9	X_{24} 20	25
3	X_{31} 5 4	X_{32} 14	X_{33} 16	X_{34} 5 18	10
	5	15	15	15	

Basic variable	(u,v) Equation	Solution
X12	$U_1 + V_2 = 2$	Set $U_1=0 \rightarrow V_2=2$
X14	$U_1 + V_4 = 11$	$U_1=0 \rightarrow V_4=11$
X22	$U_2 + V_2 = 7$	$V_2=2 \rightarrow U_2=5$
X23	$U_2 + V_3 = 9$	$U_2=5 \rightarrow V_3=4$
X34	$U_3 + V_4 = 18$	$V_4=11 \rightarrow U_3=7$
X31	$U_3 + V_1 = 4$	$U_3=7 \rightarrow V_1=-3$

Example

- Use u_i , v_j to evaluate the nonbasic variable by computing:

$U_i + v_j - c_{ij}$, for each nonbasic x_{ij}

	Demand				
Source	A	B	C	D	
1	X_{11} 10	X_{12} 5 2	X_{13} 20	X_{14} 10 11	15
2	X_{21} 12	X_{22} 10 7	X_{23} 15 9	X_{24} 20	25
3	X_{31} 5 4	X_{32} 14	X_{33} 16	X_{34} 5 18	10
	5	15	15	15	

Nonbasic variable	$U_i + V_j - c_{ij}$
X_{11}	$U_1 + V_1 - C_{11} = 0 - 3 - 10 = -13$
X_{13}	$U_1 + V_3 - C_{13} = 0 + 4 - 20 = -16$
X_{21}	$U_2 + V_1 - C_{21} = 5 - 3 - 12 = -10$
X_{32}	$U_3 + V_2 - C_{32} = 7 + 2 - 14 = -5$
X_{33}	$U_3 + V_3 - C_{33} = 7 + 4 - 16 = -5$
X_{24}	$U_2 + V_4 - C_{24} = 5 + 11 - 20 = -4$

Example

Basic	X11	X12	X13	X14	X21	X22	X23	X24	X31	X32	X33	X34
Z	-13	0	-16	0	-10	0	0	-4	0	-5	-5	0

- Since no positive , the optimal solution is

From Silo	To mill	Number truckload
1	2	5
1	4	10
2	2	10
2	3	15
3	1	5
3	4	5

- Optimal cost is \$435